

# Introduction to Bioinformatics

## RNA Structure and Transcription

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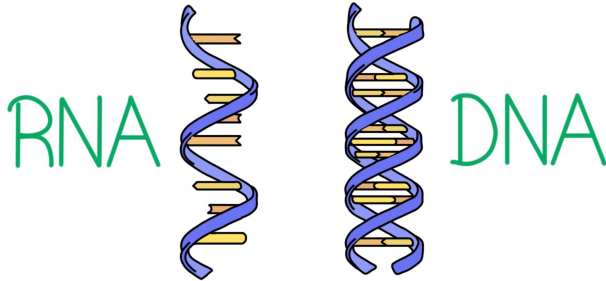
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# Presentation Overview

- ① RNA Structure and Chemistry
- ② Transcription: From DNA to RNA
- ③ Types of RNA and Gene Regulation

# The RNA World Hypothesis — Origin of Modern Biology



## Concept:

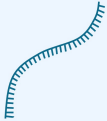
- Early life may have relied on RNA both for **information storage** and **catalysis**
- RNA molecules capable of self-replication could have preceded DNA and proteins
- Modern examples: ribozymes, ribosomal RNA catalysis

## Significance

RNA's dual ability to store genetic information and catalyze reactions supports its central role in the origin of life.

# Structured vs. Unstructured RNA Molecules

## mRNA



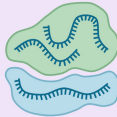
Encodes proteins

## tRNA



Carries amino acids

## rRNA



Forms the ribosome

Some RNA regions remain **unstructured**, like parts of **mRNA**.

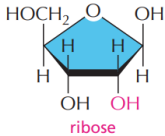
Others fold into **complex 3D structures**, such as:

- **tRNA** (adaptor in translation)
- **Ribozymes** (catalytic RNAs)
- **rRNA** (structural core of ribosome)

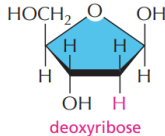
Structure reflects RNA's diverse cellular roles.

# RNA Differs from DNA in Two Key Ways

## (A) SUGAR DIFFERENCES

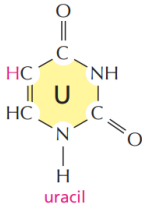


used in RNA

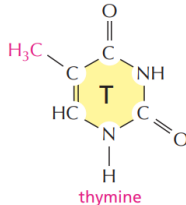


used in DNA

## (B) BASE DIFFERENCES



used in RNA



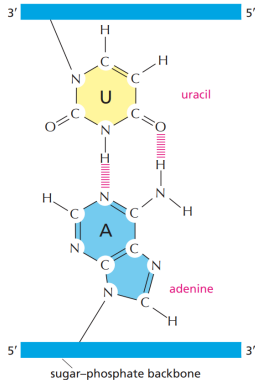
used in DNA

**1. Sugar:** RNA contains **ribose** (extra -OH group)

**2. Base:** RNA uses **uracil (U)** instead of thymine (T)

Both linked by phosphodiester bonds

# Uracil Pairs with Adenine Like Thymine



U forms hydrogen bonds with A, just like T does in DNA

U lacks the methyl group ( $-\text{CH}_3$ ) present in T

Same base-pairing rules apply to RNA

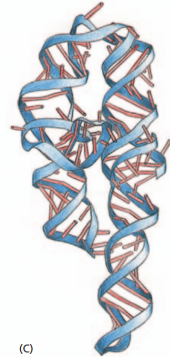
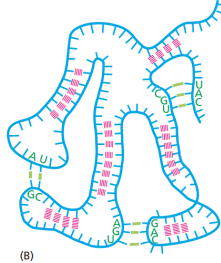
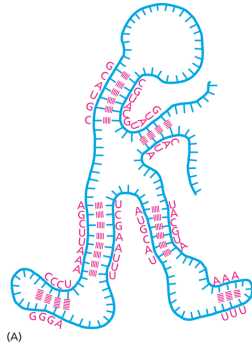
# RNA is Single-Stranded — DNA is Double-Stranded

## Key Structural Difference

- DNA: always double helix
- RNA: single-stranded → can fold into complex 3D shapes
- This allows RNA to have structural, catalytic, and regulatory roles

RNA's ability to fold enables diverse cellular functions beyond just carrying information

# RNA Folds Into Complex Structures via Base-Pairing





# RNA Secondary Structure — Key Features

## How RNA Folds

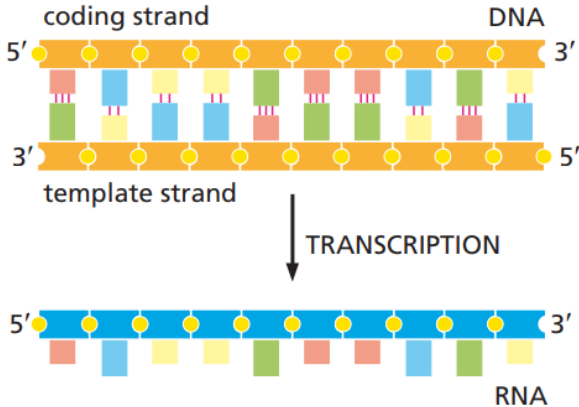
- RNA can base-pair with itself at complementary regions
- Forms hairpins, loops, and double-helical stems
- Uses conventional base pairs (G-C, A-U) and non-conventional pairs (e.g., G-U)
- 3D structure is determined by nucleotide sequence

## Functional Importance

This folding capability allows RNA to function as:

- Structural scaffold (e.g., ribosomal RNA)
- Catalytic molecule (ribozymes)
- Regulatory element (e.g., microRNA)

# Transcription: RNA Complementary to DNA



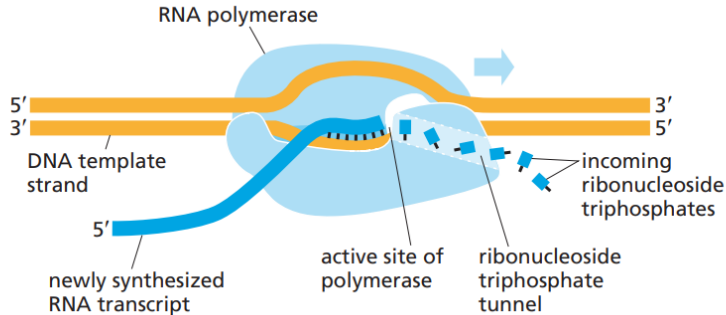
One DNA strand serves as **template**

RNA polymerase adds ribonucleotides 5'→3'

RNA sequence is complementary to template strand

RNA is single-stranded and released

# RNA Polymerase Transcribes DNA into RNA



# RNA Polymerase Mechanism

## Step-by-Step Process

- 1 **DNA Unwinding:** RNA polymerase unwinds the DNA double helix
- 2 **Template Recognition:** Uses one strand as template (read 3'→5')
- 3 **RNA Synthesis:** Adds ribonucleotides 5'→3' direction
- 4 **DNA Re-formation:** DNA double helix re-forms behind the polymerase

## Key Differences from DNA Replication

- RNA polymerase does NOT require a primer
- Only one DNA strand is used as template
- RNA remains single-stranded
- Lower fidelity:  $\sim 1$  error per  $10^4$  nucleotides

# Transcription in Prokaryotes vs. Eukaryotes

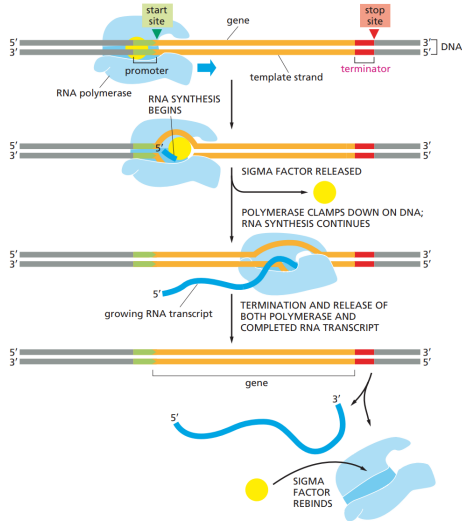
## Prokaryotes

- Occurs in cytoplasm (no nucleus)
- Transcription and translation are coupled
- Single RNA polymerase
- Simple promoters (-10 and -35 regions)
- No RNA processing steps

## Eukaryotes

- Transcription in nucleus, translation in cytoplasm
- Three RNA polymerases (I, II, III)
- Complex promoter structures (TATA box, enhancers)
- Pre-mRNA undergoes capping, splicing, and polyadenylation

# Promoters and Terminators Signal Start and Stop



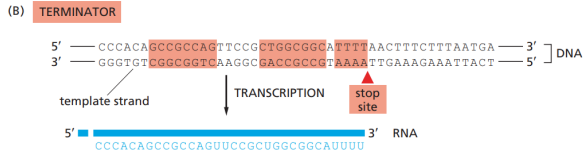
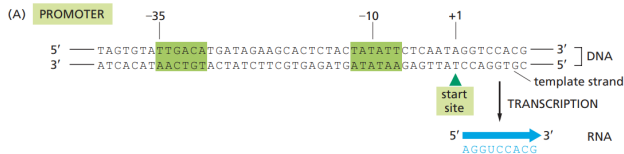
**Promoter:** RNA polymerase binds here

**Sigma factor** recognizes promoter

**Terminator:** signals transcription end

Polymerase releases RNA and DNA

# Bacterial Promoters Have Conserved Sequences

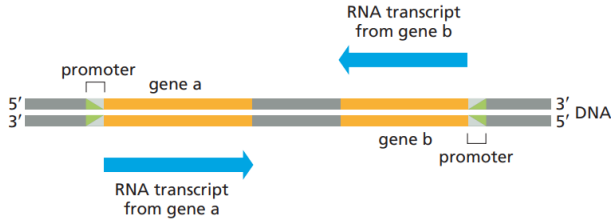


Sequences at -10 and -35 positions

Promoter polarity determines which strand is template

Only one reading frame is correct

# Promoter Orientation Determines Template Strand



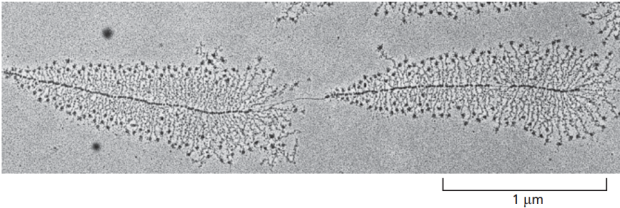
Different genes use different strands

Promoter orientation controls direction

RNA polymerase always moves 3'→5' on template



# Multiple Polymerases Transcribe Simultaneously



Many polymerases on same gene

Growing RNA transcripts visible as threads

Enables rapid mRNA production

Ribosomal proteins assemble on rRNA as it's synthesized

# Cells Produce Various Types of RNA

## Major RNA Types

- **mRNA** (messenger RNA): encodes proteins
- **rRNA** (ribosomal RNA): structural/catalytic core of ribosomes
- **tRNA** (transfer RNA): amino acid adaptors during translation
- **miRNA** (microRNA): gene regulation
- Other noncoding RNAs: splicing, telomere maintenance, etc.

## Note

Not all genes encode proteins — many produce functional RNA molecules!

# Eukaryotic Transcription Overview

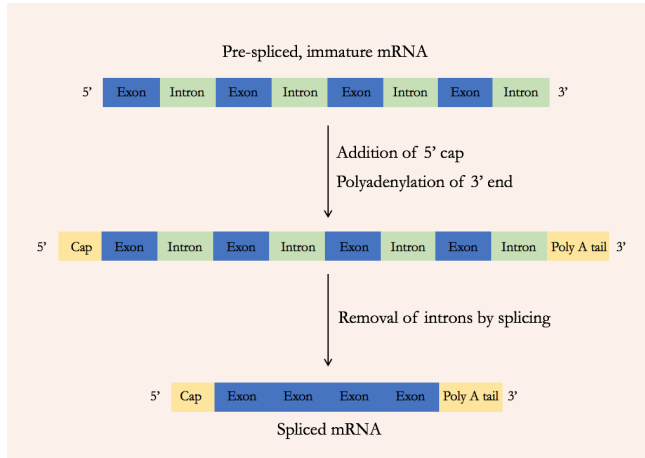
## Key Differences from Prokaryotes

- Occurs in the **nucleus**; translation occurs separately in cytoplasm.
- Uses **three RNA polymerases**: I (rRNA), II (mRNA), III (tRNA, small RNAs).
- Requires multiple **transcription factors** (no sigma factor).
- Promoters contain elements like the **TATA box**.
- Produces a **pre-mRNA** that undergoes capping, splicing, and polyadenylation.

## Summary

Eukaryotic transcription is more complex and regulated than in bacteria.

# RNA Processing in Eukaryotes: From pre-mRNA to mRNA

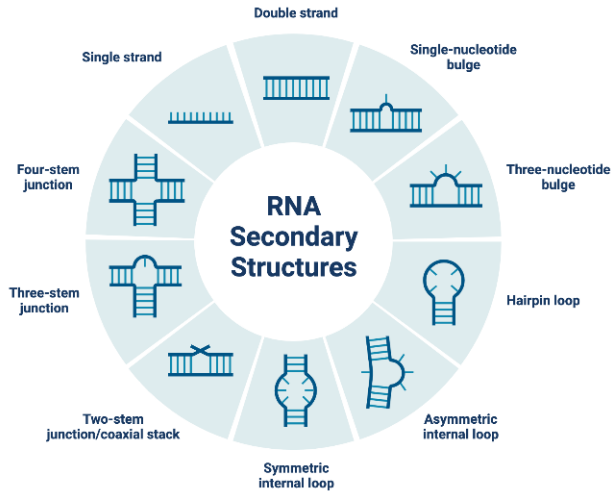


Eukaryotic RNAs undergo several modifications before leaving the nucleus:

- **5' Capping:** addition of a modified guanine nucleotide
- **Splicing:** removal of noncoding *introns* and joining of *exons*
- **3' Polyadenylation:** addition of poly(A) tail

These steps stabilize RNA and prepare it for translation in the cytoplasm.

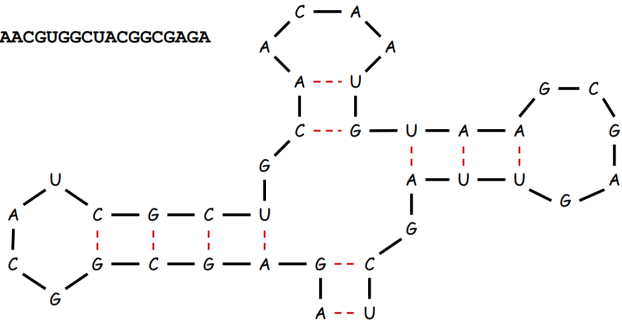
# RNA Secondary Structure



# RNA Secondary Structure: Motivation

- RNA is a single-stranded molecule made of bases **A, C, G, U**.
- It tends to **fold back and pair** with itself to form loops and stems.
- **Watson-Crick pairs:** A-U, U-A, C-G, G-C.
- Understanding secondary structure helps explain RNA function and stability.

Ex: GUCGAUUGAGCGAAUGUAACAACGUGGCUACGGCGAGA



# Problem Definition

**Given:** RNA string  $B = b_1 b_2 \dots b_n$  over the alphabet  $\{A, C, G, U\}$ .

**Find:** A set of base pairs  $S = \{(b_i, b_j)\}$  that forms a valid secondary structure.

## Constraints

- **Watson-Crick pairing:**  $(b_i, b_j)$  is valid if A-U, U-A, C-G, or G-C.
- **No sharp turns:**  $i < j - 4$  (at least 4 bases apart).
- **Non-crossing:** No  $i < k < j < l$  with both  $(b_i, b_j)$  and  $(b_k, b_l)$  in  $S$ .

**Goal:** Find  $S$  that maximizes the number of base pairs (i.e., minimizes free energy).

# Dynamic Programming Formulation

**Subproblem:** Let  $\text{OPT}(i, j)$  = maximum number of base pairs in substring  $b_i \dots b_j$ .

## Base Case

If  $i \geq j - 4$ , then  $\text{OPT}(i, j) = 0$

(no sharp turns)

## Recursive Cases

- Case 1:  $b_j$  is unpaired

$$\text{OPT}(i, j) = \text{OPT}(i, j - 1)$$

- Case 2:  $b_j$  pairs with some  $b_t$  ( $i \leq t < j - 4$ )

$$\text{OPT}(i, j) = 1 + \max_{i \leq t < j-4 \text{ s.t. } (b_t, b_j) \text{ pair}} \{\text{OPT}(i, t - 1) + \text{OPT}(t + 1, j - 1)\}$$



# Full Recurrence Relation

$$\text{OPT}(i, j) = \begin{cases} 0, & i \geq j \\ \max\left(\text{OPT}(i, j-1), 1 + \max_{\substack{i \leq t < j-4 \\ b_t \text{ pairs with } b_j}} \{\text{OPT}(i, t-1) + \text{OPT}(t+1, j-1)\}\right), & \text{else.} \end{cases}$$

- The **non-crossing** property ensures independent subproblems.
- Compute  $\text{OPT}(i, j)$  for all  $i < j$  in increasing order of  $(j - i)$ .
- $\text{OPT}(1, n)$  gives the maximum number of pairs.

# Summary and Insights

- **Goal:** Maximize number of base pairs (minimum free energy).
- **Method:** Dynamic programming over intervals.
- **Complexity:**  $O(n^3)$  time,  $O(n^2)$  space.
- **Output:**  $\text{OPT}(1, n) = \text{max pairs}$ ; can trace back to get structure.

## Key Idea

RNA folding can be viewed as an optimal non-crossing matching problem — solved efficiently using DP.

# Summary — RNA Structure and Transcription

## Key Concepts

- RNA differs from DNA by sugar (ribose) and base (uracil)
- RNA is single-stranded and folds into functional 3D shapes
- Transcription uses RNA polymerase to synthesize RNA from DNA template
- Promoters and terminators define transcription boundaries
- In eukaryotes, pre-mRNA undergoes capping, splicing, and polyadenylation

## Takeaway

RNA is more than a messenger — it plays structural, catalytic, and regulatory roles essential to gene expression and cellular control.

# References



Alberts, B., et al. (2015). *Essential Cell Biology*, 4th Edition. Garland Science, Chapter 7: From DNA to Protein.



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